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# basic education

Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

**SENIOR CERTIFICATE EXAMINATIONS/  
NATIONAL SENIOR CERTIFICATE EXAMINATIONS  
SENIORSERTIFIKAAT-EKSAMEN/  
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

**PHYSICAL SCIENCES: CHEMISTRY (P2)  
FISIESE WETENSKAPPE: CHEMIE (V2)**

**MAY/JUNE/MEI/JUNIE 2024**

**MARKING GUIDELINES/NASIENRIGLYNE**

**MARKS/PUNTE: 150**

**These marking guidelines consist of 18 pages./  
Hierdie nasienriglyne bestaan uit 18 bladsye.**

**QUESTION 1/VRAAG 1**

- 1.1 A ✓✓ (2)
- 1.2 C ✓✓ (2)
- 1.3 A ✓✓ (2)
- 1.4 D ✓✓ (2)
- 1.5 D ✓✓ (2)
- 1.6 C ✓✓ (2)
- 1.7 B ✓✓ (2)
- 1.8 A ✓✓ (2)
- 1.9 B ✓✓ (2)
- 1.10 B ✓✓ (2)
- [20]**

**QUESTION 2/VRAAG 2**

- 2.1 Organic compounds that consist of hydrogen and carbon only. ✓✓ (2 or 0)  
 Organiese verbindings wat slegs uit waterstof en koolstof bestaan. (2 of 0) (2)
- 2.2.1 C and/en E ✓ (1)
- 2.2.2 D and/en H ✓✓ (2 or/of 0) (2)
- 2.2.3 A ✓ (1)

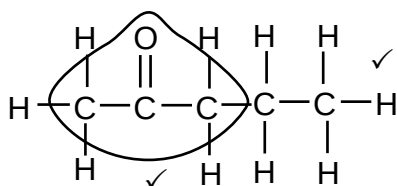
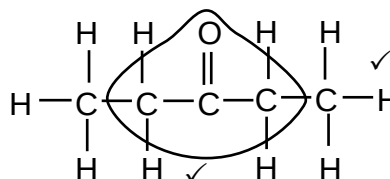
2.3  
2.3.1

**Marking criteria/Nasienkriteria:**

- Functional group. ✓  
*Funksionele groep.*
- Whole structure correct. ✓  
*Hele struktuur korrek.*

**IF/INDIEN:**

- More than one functional group/wrong functional group:  
*Meer as een funksionele groep/foutiewe funksionele groep:*  $\frac{0}{2}$
- If condensed structural formulae used/*Indien gekondenseerde struktuurformules gebruik:*  
Max/Maks.  $\frac{1}{2}$

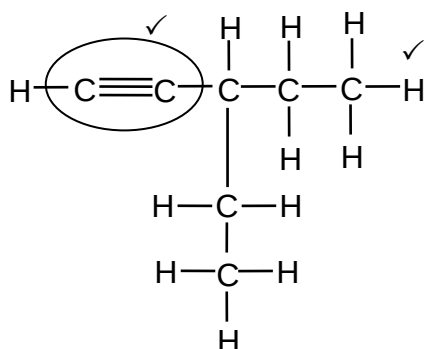
**OR/OF**

(2)

2.3.2  $C_nH_{2n+2}$  ✓

(1)

2.3.3

**Marking criteria/Nasienkriteria:**

- Functional group  $-C\equiv C-$ . ✓  
*Funksionele groep  $-C\equiv C-$ .*
- Whole structure correct. ✓  
*Hele struktuur korrek.*

**IF/INDIEN**

- More than one functional group/wrong functional group:  
*Meer as een funksionele groep/foutiewe funksionele groep:*  $\frac{0}{2}$
- If condensed structural formulae used/*Indien gekondenseerde struktuurformules gebruik:* Max/Maks.  $\frac{1}{2}$

(2)

2.4.1 3-ethylhex-3-ene ✓✓✓/3-ethyl-3-hexene/3-etielheks-3-een/3-etiel-3-hekseen

**Marking criteria:**

- Correct stem i.e. hexene. ✓
- Substituent (ethyl) correctly identified. ✓
- IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓

**Nasienkriteria:**

- Korrekte stam d.i. hekseen. ✓
- Substituent (etiel) korrek geïdentifiseer. ✓
- IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas. ✓

(3)

2.4.2 2,5-dichloro-2,4-dimethylhexane ✓✓✓/ 2,5-dichloro-2,4-dimetielsekseen

**Marking criteria:**

- Correct stem i.e. hexane. ✓
- All substituents (dichloro and dimethyl) correctly identified. ✓
- IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓

**Nasienkriteria:**

- Korrekte stam d.i. heksaan. ✓
- Alle substituenten (dichloro en dimetiel) korrek geïdentifiseer. ✓
- IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas. ✓

(3)

2.4.3 2,2-dimethylpropanal ✓/dimethylpropanal2,2-dimetielsekseen/dimetielsekseen**NOTE/NOTA:**2,2-dimethylpropan-1-al (Max/Maks:  $\frac{1}{2}$ )

(2)

2.5

**Marking criteria/Nasienkriteria:**

- Correct molecular formula:  $C_7H_{16}$  ✓  
*Korrekte molekulêre formula:  $C_7H_{16}$*
- Correct molecular formula of inorganic reactant and products. ✓  
*Korrekte molekulêre formule vir die anorganiese reaktans en produkte.*
- Balancing/Balansering ✓

**Notes/Aantekeninge:**

- Ignore double arrows and phases. *Ignoreer dubbelpyle en fases.*
- Marking rule 6.3.10/Nasienreël 6.3.10.
- If condensed structural formulae used: *Indien gekondenseerde struktuurformules gebruik:* Max/Maks.  $\frac{2}{3}$

(3)  
[22]**QUESTION 3/VRAAG 3**

3.1

**Marking criteria/Nasienkriteria**

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark. *Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.*

The underlined phrases must be in the correct context. *Die onderstreepte frases moet in die korrekte konteks wees.*

The temperature at which the vapour pressure (of a substance) equals atmospheric pressure. ✓✓

*Die temperatuur waarby die dampdruk (van die stof) gelyk is aan atmosferiese druk.*

(2)

3.2

C ✓

(1)

3.3

**Marking criteria:**

- Compare structures. ✓
- Compare the strength of intermolecular forces. ✓
- Compare the energy required to overcome intermolecular forces. ✓

**Nasienkriteria:**

- Vergelyk strukture. ✓
- Vergelyk die sterkte van intermolekulêre kragte. ✓
- Vergelyk die energie benodig om intermolekulêre kragte te oorkom. ✓

**Accept:** IMF for this exam/**Aanvaar:** IMK vir hierdie eksamen**A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>Cl /1-chlorobutane**

- **Structure:**  
Longer chain length/larger surface area (over which intermolecular forces act). ✓
- **Intermolecular forces:**  
Stronger/more intermolecular forces/Van der Waals forces/London forces/dipole-dipole forces. ✓
- **Energy:**  
More energy needed to overcome or break intermolecular forces/Van der Waals forces/dipole-dipole forces. ✓

OR

**B/CH<sub>3</sub>CH(CH<sub>3</sub>)CH<sub>2</sub>Cl/1-chloro-2-methylpropane**

- **Structure:**  
Shorter chain length / branched / compact / more spherical / smaller surface area (over which intermolecular forces act). ✓
- **Intermolecular forces:**  
Weaker/less intermolecular forces/Van der Waals forces/London forces/dipole-dipole forces. ✓
- **Energy:**  
Less energy needed to overcome or break intermolecular forces/Van der Waals forces/dipole-dipole forces. ✓

**A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>Cl /1-chlorobutaan**

- **Struktuur:**  
Langer kettinglengte/groter oppervlak (waaroor intermolekulêre kragte werk). ✓
- **Intermolekulêre kragte:**  
Sterker/meer intermolekulêre kragte/Van der Waalskragte/Londonkragte/dipool-dipoolkragte. ✓
- Meer energie benodig om intermolekulêre kragte/Van der Waalskragte/Londonkragte/dipool-dipoolkragte te oorkom/breek. ✓

OF

**B/CH<sub>3</sub>CH(CH<sub>3</sub>)CH<sub>2</sub>Cl/1-chloro-2-metielpropan**

- **Struktuur:**  
Korter kettinglengte / vertak / kompak / meer sferies / kleiner oppervlak (waaroor intermolekulêre kragte werk). ✓
- **Intermolekulêre kragte:**  
Swakker/minder intermolekulêre kragte/Van der Waalskragte/Londonkragte/dipool-dipoolkragte. ✓
- **Energie:**  
Minder energie benodig om intermolekulêre kragte/Van der Waalskragte/Londonkragte/dipool-dipoolkragte te oorkom/breek. ✓

(3)

3.4.1 75 (°C) ✓

(1)

3.4.2

**Marking criteria:**

- Compare the strength of intermolecular forces. ✓
- Compare the energy required to overcome intermolecular forces. ✓

**Nasienkriteria:**

- Vergelyk die sterkte van intermolekulêre kragte. ✓
- Vergelyk die energie benodig om intermolekulêre kragte te oorkom. ✓

- **Intermolecular forces:**

C ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ /butanol) has stronger intermolecular forces than D ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ /butanal). ✓

- **Energy:**

More energy needed to overcome or break intermolecular forces. ✓

Accept: Boiling point of C will be more (in relation to C and D/118°C vs 75°C).

OR

- **Intermolecular forces:**

D ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ /butanal) has weaker intermolecular forces than C ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ /butanol)

- **Energy:**

Less energy is needed to overcome or break intermolecular forces.

Accept: Boiling point of D will be less (in relation to C and D/118°C vs 75°C).

OR

- **Intermolecular forces:**

A ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ ) is a more polar molecule than D ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ ) increasing the intermolecular forces

- **Energy:**

More energy is needed to overcome or break intermolecular forces.

Accept: Boiling point of D will be less (in relation to A and D).

OR

- **Intermolecular forces:**

Electron density of A ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ ) is greater than D ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ ) increasing the intermolecular forces

- **Energy:**

More energy is needed to overcome or break intermolecular forces.

Accept: Boiling point of D will be less (in relation to A and D).

- **Intermolekulêre kragte:**

C ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ /butanol) het sterker intermolekulêre kragte as D ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ /butanaal). ✓

- **Meer energie benodig om intermolekulêre kragte te oorkom/breek.** ✓

Aanvaar: Kookpunt van D sal minder wees (met betrekking tot C en D)

OF

- **Intermolekulêre kragte:**

D ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ /butanaal) het swakker intermolekulêre kragte as C ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ /butanol).

- **Minder energie benodig om intermolekulêre kragte te oorkom/breek.**

Aanvaar: Kookpunt van C sal meer wees (met betrekking tot C en D)

**OF**

- **Intermolekulêre kragte:**

A ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ ) is 'n meer polêre molekule as D wat sterker intermolekulêre kragte tot gevolg het.

- Meer energie benodig om intermolekulêre kragte te oorkom/breek.

Aanvaar: Kookpunt van D sal minder wees (met betrekking tot A en D)

**OF**

- **Intermolekulêre kragte:**

Elektrondigtheid van A ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ ) is groter wat sterker intermolekulêre kragte tot gevolg het.

- Meer energie benodig om intermolekulêre kragte te oorkom/breek.

- Aanvaar: Kookpunt van D sal minder wees (met betrekking tot A en D)

(2)

3.5 Decreases/Neem af ✓

(1)

**[10]****QUESTION 4/VRAAG 4**

4.1

4.1.1 (Concentrated) sulphuric acid/ $\text{H}_2\text{SO}_4(\text{aq})$  ✓  
(Gekonsentreerde) swawelsuur

(1)

4.1.2 Esterification / Condensation ✓ / Verestering / Esterifikasie / Kondensasie

(1)

4.1.3 **ANY TWO/ENIGE TWEE:**

- Alcohol/methanol/reactant is flammable/catches fire easily. ✓

Alkohol/metanol/reaktans is vlambaar/slaan maklik aan die brand.

- To heat evenly/A steady/controlled/gradual increase in temperature. ✓  
Om eweredig/gekontroleerd/gelydelik te verhit/'n Eweredige toename in temperatuur.

- Alcohol/methanol will evaporate too quickly/is volatile.

Alkohol/metanol sal te vinnig verdamp/is vlugtig.

(2)



4.1.4

**Marking criteria:**

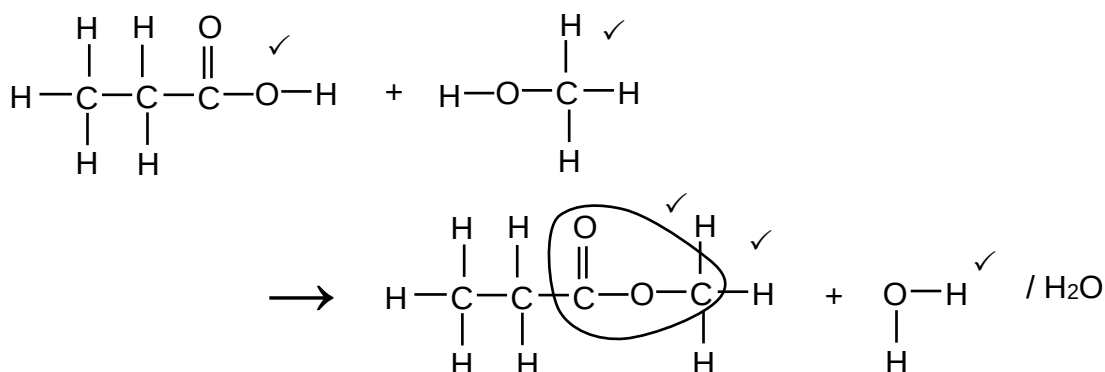
- Whole structural formula correct for propanoic acid. ✓
- Whole structural formula correct for methanol. ✓
- Functional group of ester correct. ✓
- Whole structural formula of ester correct. ✓
- H<sub>2</sub>O ✓

**Nasienkriteria:**

- Hele struktuurformule vir propaansuur korrek. ✓
- Hele struktuurformule vir metanol korrek. ✓
- Funksionele groep van ester korrek. ✓
- Hele struktuurformule van ester korrek. ✓
- H<sub>2</sub>O ✓

**IF/INDIEN**

- Any error e.g. omission of all H atoms, condensed or semi structural formula/Enige fout bv. weglating van alle H-atome, gekondenseerde of semi-struktuurformule: Max/Maks.  $\frac{2}{5}$  (Functional group, H<sub>2</sub>O/Funksionele groep, H<sub>2</sub>O)
- Any additional reactants or products /Enige addisionele reaktanse of produkte: Subtract 1 mark./Trek 1 punt af.
- Molecular formulae used:/Molekulêre formule gebruik: Max/Maks.  $\frac{1}{5}$  (water)
- No arrows: The first two structures given are considered as reactants and can be marked/Geen pyltjie: die eerste twee strukture geskryf, word beskou as reaktanse en kan gemerk word.



(5)

4.1.5 Methyl ✓propanoate ✓/Metielpropanoaat

(2)

4.2.1 Hydrogen/H<sub>2</sub> ✓/Waterstof(gas)

(1)

4.2.2 3,3-dimethyl✓but-1-ene✓/3,3-dimethyl-1-butene  
3,3-dimetiel but-1-een/3,3-dimetiel-1-buteen

(2)

4.2.3 elimination **OR** dehydrohalogenation ✓ eliminاسie **OF** dehidrohalogenering

(1)

4.2.4 H<sub>2</sub>SO<sub>4</sub>/H<sub>3</sub>PO<sub>4</sub> **OR/OF** Sulphuric acid/Phosphoric acid ✓  
Swawelsuur/Fosforsuur

(1)

4.2.5 3,3-dimethyl✓butan-2-ol✓/3,3-dimethyl-2-butanol  
3,3-dimetiel butan-2-ol/3,3-dimetiel-2-butanol

(2)

4.2.6 Addition/hydration ✓ Addisie/hidrasie

(1)

4.2.7 Secondary ✓/Sekondêr

(1)

**[20]**

**QUESTION 5/VRAAG 5**

## 5.1.1 Exothermic/Eksotermies ✓

Lower (potential) energy of the products than reactants.  $\Delta H < 0$  /  $\Delta H$  negative /  $\Delta H = -121,7$  kJ / More energy is released than absorbed. ✓

*Laer (potensiële) energie van produkte as die reaktanse.  $\Delta H < 0$  /  $\Delta H$  negatief /  $\Delta H = -121,7$  kJ / Meer energie word afgegee as wat opgeneem is.*

(2)

5.1.2 (The number of) particles with sufficient/enough (kinetic) energy (with a catalyst) OR  $E_K \geq E_A$  (which can undergo effective collisions.) ✓

*(Die hoeveelheid) deeltjies met genoeg/voldoende (kinetiese) energie (met 'n katalisator) OF  $E_K \geq E_A$  (om effektiewe botsings te ondergaan).*

(1)

5.1.3  $240,8 - 208,2 = 32,6$  (kJ) ✓

(2)

**IF:** only answer award 2 marks / **INDIEN:** slegs antwoord gee 2 punte

## 5.2

## 5.2.1 Decreases/Afneem ✓

(1)

## 5.2.2 Remains the same/Bly dieselfde ✓

(1)

## 5.2.3 Remains the same/Bly dieselfde ✓

(1)

5.3.1 Concentration (of sulphuric acid/ $H_2SO_4(aq)$ ) / Konsentrasie (van swawelsuur) ✓

(1)

5.3.2 • More ( $H_2SO_4$ ) particles per unit volume. ✓

• More effective collisions per unit time. / Higher frequency of effective collisions. ✓

• Higher reaction rate. ✓

**OR**

• Less ( $H_2SO_4$ ) particles per unit volume. ✓

• Less effective collisions per unit time. / Lower frequency of effective collisions. ✓

• Lower reaction rate ✓

• Meer ( $H_2SO_4$ ) deeltjies per eenheid volume. ✓

• Meer effektiewe botsings per eenheidtyd. / Hoër frekwensie van effektiewe botsings. ✓

• Hoër reaksietempo. ✓

**OF**

• Minder ( $H_2SO_4$ )-deeltjies per eenheid volume. ✓

• Minder effektiewe botsings per eenheidtyd. / Laer frekwensie van effektiewe botsings. ✓

• Laer reaksietempo. ✓

(3)

5.3.3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Marking criteria:</b></p> <p>(a) Substitute <math>(2,6)(60)(40) \text{ cm}^3</math> OR <math>(156)(40)</math> in rate formula ✓</p> <p>(b) Substitute <math>27\,000 \text{ cm}^3 / 27 \text{ dm}^3</math> and volume in <math>n(\text{H}_2) = \frac{V}{V_m}</math> ✓</p> <p>(c) USE mole ratio <math>n(\text{Al}) = \frac{2}{3}n(\text{H}_2)</math> ✓</p> <p>(d) Substitution 27 and reacting mole in <math>n(\text{Al}) = \frac{m}{M}</math> ✓</p> <p>(e) Substitution of <math>\frac{4,05}{5}(100)</math> ✓</p> <p>(f) Final answer: 83,2 % ✓<br/>Range: 81 – 83,3 %</p>                                                                                                                                                                                                                                                                                                                                                              | <p><b>Nasienkriteria:</b></p> <p>(a) Vervang <math>2,6(60)(40) \text{ cm}^3</math> OF <math>(156)(40)</math> in tempo formule ✓</p> <p>(b) Vervang <math>27\,000 \text{ cm}^3 / 27 \text{ dm}^3</math> en volume in <math>n(\text{H}_2) = \frac{V}{V_m}</math> ✓</p> <p>(c) GEBRUIK molverhouding <math>n(\text{Al}) = \frac{2}{3}n(\text{H}_2)</math> ✓</p> <p>(d) Vervang 27 en mol gereageer in <math>n(\text{Al}) = \frac{m}{M}</math> ✓</p> <p>(e) Vervang van <math>\frac{4,05}{5}(100)</math> ✓</p> <p>(f) Finale antwoord: 81 % ✓<br/>Gebied: 81 – 83,3 %</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>OPTION 1/OPSIE 1:</b></p> <p>Rate/Tempo = <math>\frac{\Delta V_{\text{H}_2}}{\Delta t}</math></p> <p><math>40 = \frac{\Delta V_{\text{H}_2}}{2,6(60)} \quad \checkmark \text{ (a)}</math></p> <p><math>V(\text{H}_2) = 6\,240 \text{ cm}^3</math></p> <p><math>n(\text{H}_2) = \frac{V}{V_m}</math></p> <p><math>= \frac{6\,240}{27\,000} \quad \checkmark \text{ (b)}</math></p> <p><math>= 0,23 \text{ mol}</math></p> <p><math>n(\text{Al}) = \frac{2}{3} n(\text{H}_2)</math></p> <p><math>n(\text{Al}) = \frac{2}{3} (0,23) \quad \checkmark \text{ (c)}</math></p> <p><math>= 0,15 \text{ mol}</math></p> <p><math>n(\text{Al}) = \frac{m}{M}</math></p> <p><math>0,15 = \frac{m}{27} \quad \checkmark \text{ (d)}</math></p> <p><math>m = 4,05 \text{ g}</math></p> <p><math>\% \text{ purity/suiwerheid} = \frac{4,05}{5}(100) \quad \checkmark \text{ (e)}</math></p> <p><math>= 81 \% \quad \checkmark \text{ (f)}</math></p> | <p><b>OPTION 2/OPSIE 2:</b></p> <p>rate <math>\text{H}_2 = 40 \text{ cm}^3 \cdot \text{s}^{-1}</math></p> <p>Rate in <math>n(\text{H}_2) = \frac{V}{V_m}</math></p> <p><math>= \frac{40}{27\,000} \quad \checkmark \text{ (b)}</math></p> <p><math>= 0,00148 \text{ mol} \cdot \text{s}^{-1}</math></p> <p>Rate <math>(\text{Al}) = \frac{2}{3} n(\text{H}_2)</math></p> <p><math>= \frac{2}{3} (0,00148) \quad \checkmark \text{ (c)}</math></p> <p><math>= 9,88 \times 10^{-4} \text{ mol} \cdot \text{s}^{-1}</math></p> <p><math>n(\text{Al}) = \frac{m}{M}</math></p> <p><math>9,88 \times 10^{-4} = \frac{m}{27} \quad \checkmark \text{ (d)}</math></p> <p><math>m = 0,0267 \text{ g} \cdot \text{s}^{-1}</math></p> <p>Rate/Tempo = <math>\frac{\Delta m_{\text{Al}}}{\Delta t}</math></p> <p><math>0,0267 = \frac{\Delta m_{\text{Al}}}{2,6(60)}</math></p> <p><math>m(\text{Al}) = 4,16 \text{ g}</math></p> <p><math>\% \text{ purity/suiwerheid} = \frac{4,16}{5}(100) \quad \checkmark \text{ (e)}</math></p> <p><math>= 83,2 \% \quad \checkmark \text{ (f)}</math></p> |

(6)  
[18]

**QUESTION 6/VRAAG 6**

6.1

**Marking criteria/Nasienkriteria:**

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark./Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The underlined phrases must be in the correct context. / Die onderstreepte frases moet in die korrekte konteks wees.

When the equilibrium in a closed system is disturbed, the system will re-instate a new equilibrium by favouring the reaction that will cancel/oppose the disturbance. ✓✓

Wanneer die ewewig in 'n geslote sisteem versteur word, sal die sisteem 'n nuwe ewewig instel deur die reaksie te bevoordeel wat die versteuring kanselleer/teenwerk.

IF “isolated” system -1/**INDIEN:** “geïsoleerde” sisteem -1)

(2)

6.2

(Chemical) equilibrium/Concentrations of reactants and products remain constant./Rate of the forward and reverse reactions are equal. ✓

(Chemiese) ewewig/Konsentrasies van reaktante en produkte bly konstant./Tempo van voorwaartse en terugwaartse reaksie is gelyk.

(1)

6.3

Exothermic/Eksotermies ✓

(1)

6.4

- With an increase in temperature the endothermic reaction is favoured. ✓
- The reverse reaction is favoured./ Equilibrium shifts to the left. / Reactants /  $[P_2Q]$  increases OR Products /  $[PQ_2]$  decreases ✓
- 'n Toename in temperatuur bevoordeel die endotermiese reaksie.
- Die terugwaartse reaksie word bevoordeel./ Ewewig skuif na links. / Reaktante /  $[P_2Q]$  neem toe OF Produkte /  $[PQ_2]$  neem af

(2)

6.5

Less than/Kleiner as ✓

(1)

6.6

**CALCULATIONS USING CONCENTRATION****Marking criteria:**

- (a) Correct  $K_c$  expression (formulae in square brackets). ✓✓  
(If solid is included deduct 1 mark)
- (b) Substitute 0,49 into  $K_c$  expression. ✓
- (c) Substitute equilibrium concentration (0,35) into correct  $K_c$  expression. ✓
- (d) Change in concentration/mole ✓
- (e) **USE** ratio:  $P_2Q : 2PQ_2 = 1 : 2$  ✓
- (f) Substitute  $2 \text{ dm}^3$  in  $n = cV$ . ✓
- (g) Final answer = 0,85 (mol) OR 1,11 (mol) OR 3,09 (mol) ✓

**Nasienkriteria:**

- (a) Korrekte  $K_c$  uitdrukking (formules in vierkantige hakies). ✓✓  
(Indien vastestof inervang is, trek 1 punt af)
- (b) Vervang 0,49 in  $K_c$ -uitdrukking. ✓
- (c) Vervang ewewigkonsentrasie (0,35) in korrekte  $K_c$ -uitdrukking. ✓
- (d) Verandering in konsentrasie/mol ✓
- (e) **GEBRUIK** verhouding:  $P_2Q : PQ_2 = 1 : 2$  ✓
- (f) Vervang  $2 \text{ dm}^3$  in  $n = cV$ . ✓
- (g) Finale antwoord = 0,85 (mol) OF 1,11 (mol) OF 3,09 (mol) ✓

**OPTION 1/OPSIE 1:**

|                                                                                                                                                    | $P_2Q$         | $PQ_2$ |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------|
| Initial concentration ( $\text{mol} \cdot \text{dm}^{-3}$ )<br><i>Aanvangskonsentrasie (<math>\text{mol} \cdot \text{dm}^{-3}</math>)</i>          | x              | 0      |
| Change in concentration ( $\text{mol} \cdot \text{dm}^{-3}$ )<br><i>Verandering in konsentrasie (<math>\text{mol} \cdot \text{dm}^{-3}</math>)</i> | 0,175 ✓(e)     | 0,35   |
| Equilibrium concentration ( $\text{mol} \cdot \text{dm}^{-3}$ )<br><i>Ewewigskonsentrasie (<math>\text{mol} \cdot \text{dm}^{-3}</math>)</i>       | ✓(d) x - 0,175 | 0,35   |

$$K_c = \frac{[PQ_2]^2}{[P_2Q]} \quad \checkmark\checkmark \text{ (a)}$$

$$0,49 \quad \checkmark \text{ (b)} = \frac{(0,35)^2 \quad \checkmark \text{ (c)}}{(x - 0,175)}$$

$$x = 0,425 \text{ mol} \cdot \text{dm}^{-3}$$

$$n(P_2Q) = cV \quad \swarrow$$

$$= 0,425 \times 2 \quad \checkmark \text{ (f)}$$

$$= 0,85 \text{ mol} \quad \checkmark \text{ (g)}$$

No  $K_c$  expression, correct substitution/Geen  $K_c$ -uitdrukking, korrekte substitusie: Max./Maks.  $\frac{6}{8}$

Wrong  $K_c$  expression/  
Verkeerde  $K_c$ -uitdrukking: Max./Maks.  $\frac{5}{8}$

**OPTION2/OPSIE 2:**

$$K_c = \frac{[PQ_2]^2}{[P_2Q]}$$

$$0,49 \quad \checkmark (b) = \frac{(0,35)^2 \quad \checkmark (c)}{P_2Q}$$

$$P_2Q = 0,25 \text{ mol} \cdot \text{dm}^{-3}$$

No  $K_c$  expression, correct substitution/Geen  $K_c$ -uitdrukking, korrekte substitusie: Max./Maks.  $\frac{6}{8}$

Wrong  $K_c$  expression/  
Verkeerde  $K_c$ -uitdrukking: Max./Maks.  $\frac{5}{8}$

|                                                                                                                                    | $P_2Q$                 | $PQ_2$                |
|------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------|
| Initial concentration ( $\text{mol} \cdot \text{dm}^{-3}$ )<br>Aanvangskonsentrasie ( $\text{mol} \cdot \text{dm}^{-3}$ )          | $\checkmark (d)$ 0,425 | 0                     |
| Change in concentration ( $\text{mol} \cdot \text{dm}^{-3}$ )<br>Verandering in konsentrasie ( $\text{mol} \cdot \text{dm}^{-3}$ ) | -0,175                 | $\checkmark (e)$ 0,35 |
| Equilibrium concentration ( $\text{mol} \cdot \text{dm}^{-3}$ )<br>Ewewigskonsentrasie ( $\text{mol} \cdot \text{dm}^{-3}$ )       | 0,25                   | 0,35                  |

$$n(P_2Q) = cV$$

$$= 0,425(2) \quad \checkmark (f)$$

$$= 0,85 \text{ mol} \quad \checkmark (g)$$

**CALCULATIONS USING NUMBER OF MOLES****OPTION 3/OPSIE 3:**

|                                                                                                                              | $P_2Q$                                | $PQ_2$ |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------|
| Initial quantity (mol)<br>Aanvangshoeveelheid (mol)                                                                          | x                                     | 0      |
| Change (mol)<br>Verandering (mol)                                                                                            | $\checkmark (e)$ 0,35                 | 0,7    |
| Quantity at equilibrium (mol)<br>Hoeveelheid by ewewig (mol)                                                                 | $\checkmark (d)$ $x - 0,35$           | 0,7    |
| Equilibrium concentration ( $\text{mol} \cdot \text{dm}^{-3}$ )<br>Ewewigskonsentrasie ( $\text{mol} \cdot \text{dm}^{-3}$ ) | $\checkmark (f)$ $\frac{x - 0,35}{2}$ | 0,35   |

$$K_c = \frac{[PQ_2]^2}{[P_2Q]}$$

$$0,49 \quad \checkmark (b) = \frac{(0,35)^2 \quad \checkmark (c)}{\left(\frac{x - 0,35}{2}\right)}$$

$$x = 0,85 \text{ mol} \quad \checkmark (g)$$

No  $K_c$  expression, correct substitution/Geen  $K_c$ -uitdrukking, korrekte substitusie: Max./Maks.  $\frac{6}{8}$

Wrong  $K_c$  expression/  
Verkeerde  $K_c$ -uitdrukking: Max./Maks.  $\frac{5}{8}$

**OPTION 4/OPSIE 4:**

$$K_c = \frac{[PQ_2]^2}{[P_2Q]} \quad \checkmark \checkmark \text{ (a)}$$

$$0,49 \checkmark \text{ (b)} = \frac{(0,35)^2}{[P_2Q]} \quad \checkmark \text{ (c)}$$

$$[P_2Q] = 0,25 \text{ mol} \cdot \text{dm}^{-3}$$

Wrong  $K_c$  expression/Verkeerde  $K_c$ -uitdrukking: Max./Maks.  $\frac{5}{8}$ No  $K_c$  expression, correct substitution/Geen  $K_c$ -uitdrukking, korrekte substitusie: Max./Maks.  $\frac{6}{8}$ 

|                                                                                                                              | $P_2Q$                         | $PQ_2$                       |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------------|
| Initial quantity (mol)<br>Aanvangshoeveelheid (mol)                                                                          | $\checkmark \text{ (g)}$ 0,85  | 0                            |
| Change (mol)<br>Verandering (mol)                                                                                            | $\checkmark \text{ (e)}$ -0,35 | 0,7 $\checkmark \text{ (d)}$ |
| Quantity at equilibrium (mol)<br>Hoeveelheid by ewewig (mol)                                                                 | 0,5 $\checkmark \text{ (f)}$   | 0,7                          |
| Equilibrium concentration ( $\text{mol} \cdot \text{dm}^{-3}$ )<br>Ewewigskonsentrasie ( $\text{mol} \cdot \text{dm}^{-3}$ ) | 0,25                           | 0,35                         |

- 6.7 Pressure was decreased/volume of the container was increased.  $\checkmark$   
 Druk is verlaag/volume van die houer is vergroot.

(1)

- 6.8
- Favours the reaction that increases the number of moles (of gas)  $\checkmark$ /Bevoordeel die reaksie wat aantal mol (gas) laat toeneem
  - $[P_2Q]$  increased/neem toe  $\checkmark$

(2)

**[18]**

**QUESTION 7/VRAAG 7**

7.1

**Marking criteria:**

- Any formula  $c = \frac{m}{MV}$  or  $n = \frac{m}{M}$  or  $c = \frac{n}{V}$  ✓
- Substitute 10, 106 and 0,7 into formula ✓
- Final answer:  $0,13 \text{ mol} \cdot \text{dm}^{-3}$  ✓

**Nasienkriteria:**

- Enige formule  $c = \frac{m}{MV}$  of  $n = \frac{m}{M}$  of  $c = \frac{n}{V}$  ✓
- Vervang 10, 106 and 0,7 in formula ✓
- Finale antwoord:  $0,13 \text{ mol} \cdot \text{dm}^{-3}$  ✓

7.1.1

**OPTION 1/OPSIE 1:**

$$c = \frac{m}{MV} \quad \checkmark$$

$$= \frac{10}{(106)(0,7)} \quad \checkmark$$

$$= 0,13 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

**OPTION 2/OPSIE 2:**

$$n = \frac{m}{M} \quad \text{Any one/Enige een } \checkmark$$

$$= \frac{10}{106} \quad \checkmark$$

$$= 0,09$$

$$c = \frac{n}{V}$$

$$= \frac{0,09}{0,7} \quad \checkmark$$

$$= 0,13 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

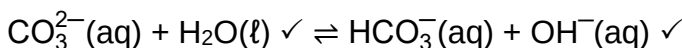
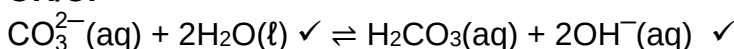
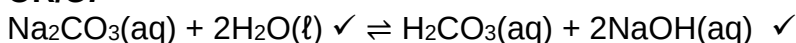
(3)

7.1.2

Greater than/Groter as ✓

(1)

7.1.3

**OR/OF****OR/OF****OR/OF****Marking criteria/Nasienkriteria:**

- Reactants ✓ Products ✓  
Reaktanse ✓ Produkte ✓
- Ignore/Ignoreer → and phases/en fases
- Marking rule 6.3.10/Nasienreël 6.3.10

(2)

7.1.4

⊖ P ✓

(Titration of) weak base and a strong acid./The equivalence point is lower than pH 7. ✓

(Titrasië van) 'n swak basis en 'n sterk suur./ Die ekwivalente punt is laer as 'n pH van 7.

(2)

7.2

7.2.1

Dilute acid contains small amount/number of moles of acid in proportion to the volume of water. ✓✓ **(2 orlof 0)**

Verdunde sure bevat 'n klein hoeveelheid/getal mol suur in verhouding met die volume water.

(2)



7.2.2

| <b>Marking criteria:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Nasienkriteria:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>(a) USE of ratio:</b><br/> <math>n(\text{KOH})_{\text{reacted}} = 2n(\text{H}_2\text{SO}_4)_{\text{reacted}}</math><br/> <math>[\text{KOH}]_{\text{reacted}} = 2n[\text{H}_2\text{SO}_4]_{\text{reacted}}</math> ✓</p> <p><b>(b) Subtract:</b> <math>n(\text{KOH})_{\text{initial}} - n(\text{KOH})_{\text{reacted}}</math><br/> <math>[\text{KOH}]_{\text{initial}} - [\text{KOH}]_{\text{reacted}}</math> ✓✓</p> <p><b>(c) Divide n by 0,20 dm<sup>3</sup> in c = <math>\frac{n}{V}</math></b> ✓</p> <p><b>(d) Either formulae:</b> <math>\text{pH} = -\log[\text{H}_3\text{O}^+]</math> /<br/> <math>\text{pH} = -\log[\text{H}^+] / \text{pOH} = -\log[\text{OH}^-]</math> AND<br/> <math>[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}</math> /<br/> <math>\text{pH} + \text{pOH} = 14</math> ✓</p> <p><b>(e) Substitute calculated <math>[\text{OH}^-]</math> in <math>[\text{H}_3\text{O}^+][\text{OH}^-]</math> / in <math>\text{pOH} = -\log[\text{OH}^-]</math></b> ✓</p> <p><b>(f) Substitute calculated <math>[\text{H}_3\text{O}^+]</math> in pH formula/ <math>\text{pOH}</math> in <math>\text{pH} + \text{pOH} = 14</math></b> ✓</p> <p><b>(g) Final answer: 12,3</b> ✓</p> | <p><b>(a) GEBRUIK verhouding:</b><br/> <math>n(\text{KOH})_{\text{gereageer}} = 2n(\text{H}_2\text{SO}_4)_{\text{gereageer}}</math><br/> <math>[\text{KOH}]_{\text{gereageer}} = 2n[\text{H}_2\text{SO}_4]_{\text{gereageer}}</math> ✓</p> <p><b>(b) Aftrek:</b> <math>n(\text{KOH})_{\text{aanvanklik}} - n(\text{KOH})_{\text{gereageer}}</math><br/> <math>[\text{KOH}]_{\text{aanvanklik}} - [\text{KOH}]_{\text{gereageer}}</math> ✓✓</p> <p><b>(c) Deel n deur 0,20 dm<sup>3</sup> in c = <math>\frac{n}{V}</math></b> ✓</p> <p><b>(d) Enige een v formules:</b> <math>\text{pH} = -\log[\text{H}_3\text{O}^+]</math> /<br/> <math>\text{pH} = -\log[\text{H}^+] / \text{pOH} = -\log[\text{OH}^-]</math> EN<br/> <math>[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}</math> /<br/> <math>\text{pH} + \text{pOH} = 14</math> ✓</p> <p><b>(e) Vervang berekende <math>[\text{OH}^-]</math> in <math>[\text{H}_3\text{O}^+][\text{OH}^-]</math> / in <math>\text{pOH} = -\log[\text{OH}^-]</math></b> ✓</p> <p><b>(f) Vervang berekende <math>[\text{H}_3\text{O}^+]</math> in pH formule/ <math>\text{pOH}</math> in <math>\text{pH} + \text{pOH} = 14</math></b> ✓</p> <p><b>(g) Finale antwoord: 12,3</b> ✓</p> |
| <p><b>OPTION 1/OPSIE 1:</b><br/> <math>n(\text{KOH})_{\text{reacted}} = 2n(\text{H}_2\text{SO}_4)_{\text{reacted}}</math><br/> <math>= 2(0,01)</math> ✓ (a)<br/> <math>= 0,02</math></p> <p><math>n(\text{KOH})_{\text{excess}} = 0,024 - 0,02</math> ✓✓ (b)<br/> <math>= 0,004 \text{ mol}</math></p> <p><math>[\text{OH}^-] = \frac{n}{V}</math><br/> <math>= \frac{0,004}{0,20}</math> ✓ (c)<br/> <math>= 0,02 \text{ mol} \cdot \text{dm}^{-3}</math></p> <p><math>[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}</math><br/> <math>[\text{H}_3\text{O}^+] (0,02) = 1 \times 10^{-14}</math> ✓ (e)<br/> <math>[\text{H}_3\text{O}^+] = 5 \times 10^{-13} \text{ mol} \cdot \text{dm}^{-3}</math></p> <p><math>\text{pH} = -\log[\text{H}_3\text{O}^+]</math><br/> <math>= -\log(5 \times 10^{-13})</math> ✓ (f)<br/> <math>= 12,3</math> ✓ (g)</p> <p>Either/<br/>Enige een ✓ (d)</p>                                                                                                                                                                                                                                                                                                          | <p><b>OPTION 2/OPSIE 2:</b><br/> <math>[\text{KOH}] = \frac{n}{V}</math><br/> <math>= \frac{0,024}{0,20}</math><br/> <math>= 0,12 \text{ mol} \cdot \text{dm}^{-3}</math> ✓ (c)</p> <p><math>[\text{H}_2\text{SO}_4] = \frac{n}{V}</math><br/> <math>= \frac{0,01}{0,20}</math><br/> <math>= 0,05 \text{ mol} \cdot \text{dm}^{-3}</math></p> <p><math>[\text{KOH}]_{\text{reacted}} = 2[\text{H}_2\text{SO}_4]_{\text{reacted}}</math><br/> <math>= 2(0,05)</math> ✓ (a)<br/> <math>= 0,1 \text{ mol} \cdot \text{dm}^{-3}</math></p> <p><math>[\text{KOH}]_{\text{excess}} = 0,12 - 0,1</math> ✓✓ (b)<br/> <math>= 0,02 \text{ mol} \cdot \text{dm}^{-3}</math></p> <p><math>[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}</math><br/> <math>[\text{H}_3\text{O}^+] (0,02) = 1 \times 10^{-14}</math> ✓ (e)<br/> <math>[\text{H}_3\text{O}^+] = 5 \times 10^{-13} \text{ mol} \cdot \text{dm}^{-3}</math></p> <p><math>\text{pH} = -\log[\text{H}_3\text{O}^+]</math><br/> <math>= -\log(5 \times 10^{-13})</math> ✓ (f)<br/> <math>= 12,3</math> ✓ (g)</p> <p>Both/<br/>Beide ✓ (d)</p>                                                                                                                             |
| <p><b>OPTION 3/OPSIE 3</b><br/> <math>\text{pOH} = -\log[\text{OH}^-]</math><br/> <math>\text{pOH} = -\log(0,02)</math> ✓ (e)<br/> <math>\text{pOH} = 1,7</math></p> <p><math>\text{pH} + \text{pOH} = 14</math><br/> <math>\text{pH} + 1,7 = 14</math> ✓ (f)<br/> <math>\text{pH} = 12,3</math> ✓ (g)</p> <p>Any one/Enige een ✓ (d)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

(8)  
[18]

**QUESTION 8/VRAAG 8**

8.1 Aluminium/Al ✓ (1)

8.2 0,325 (mol·dm<sup>-3</sup>) ✓✓  
Range/Gebied: 0,32 – 0,33 (mol·dm<sup>-3</sup>) (2)

8.3 Decreases / Neem af ✓  
M<sup>2+</sup> is reduced/ M<sup>2+</sup> used up/M<sup>2+</sup> is the oxidising agent. ✓ (2)  
M<sup>2+</sup> word gereduseer/ M<sup>2+</sup> opgebruik/M<sup>2+</sup> is die oksideermiddel.

8.4 M ✓ (1)

8.5

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>OPTION 1/OPTION 1</b></p> <p><math>E_{\text{cell}}^{\theta} = E_{\text{reduction}}^{\theta} - E_{\text{oxidation}}^{\theta}</math> ✓</p> <p><math>2 \checkmark \checkmark = E_{\text{cathode}}^{\theta} - (-1,66)</math> ✓</p> <p><math>E_{\text{cathode}}^{\theta} = 0,34 \text{ (V)}</math> ✓</p> <p>M is copper/Cu/koper ✓</p>                                                                                                                                                                                                                                       | <p><b>NOTE/LET WEL</b></p> <ul style="list-style-type: none"> <li>Accept any other correct formula from the data sheet. /Aanvaar enige ander korrekte formule vanaf gegewensblad.</li> <li>Any other formula using unconventional abbreviations, e.g. <math>E_{\text{cell}}^{\circ} = E_{\text{OA}}^{\circ} - E_{\text{RA}}^{\circ}</math> followed by correct substitutions: /Enige ander formule wat onkonvensionele afkortings gebruik, bv. <math>E_{\text{sel}}^{\circ} = E_{\text{OM}}^{\circ} - E_{\text{RM}}^{\circ}</math> gevolg deur korrekte vervangings <sup>5</sup>/<sub>6</sub></li> </ul> |
| <p><b>OPTION 2/OPSIE 2</b></p> <p> <math>\begin{cases} \text{M}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{M}(\text{aq}) &amp; E = +x \text{ V} \\ \text{Al}(\text{s}) \rightarrow \text{Al}^{3+}(\text{aq}) + 3\text{e}^{-} &amp; E = +1,66 \text{ V } \checkmark \end{cases}</math> </p> <p><math>2\text{Al}(\text{s}) + 3\text{M}^{2+}(\text{aq}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{M}(\text{s})</math> <math>E = 2,00 \text{ (V)} \checkmark \checkmark</math></p> <p><math>x = 0,34 \text{ (V)} \checkmark</math></p> <p>M is copper/Cu/koper ✓</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

(6)

8.6.1 Magnesium/Mg ✓ (1)

8.6.2 Al<sup>3+</sup> is a stronger oxidising agent than Mg<sup>2+</sup> ✓, therefore, Mg will be oxidised (to Mg<sup>2+</sup>). /  
Mg<sup>2+</sup> is a weaker oxidising agent than Al<sup>3+</sup> ✓, therefore, Mg will be oxidised (to Mg<sup>2+</sup>).

*Al<sup>3+</sup> is 'n sterker oksideermiddel as Mg<sup>2+</sup>, daarom sal Mg geoksideer word (tot Mg<sup>2+</sup>). /*

*Mg<sup>2+</sup> is 'n swakker oksideermiddel as Al<sup>3+</sup>, daarom sal Mg geoksideer word (tot Mg<sup>2+</sup>).*

(2)  
[15]

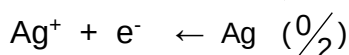
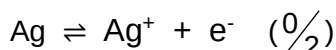
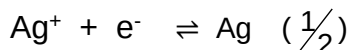
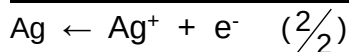
### QUESTION 9/VRAAG 9

9.1 Electrical to chemical (energy)/Elektriese na chemiese (energie) ✓ (1)

9.2 P ✓ (1)

9.3  $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$  ✓✓

**Marking criteria/Nasienkriteria:**



Ignore if charge omitted on electron./Ignoreer indien lading weggelaat op elektron.

(2)

9.4

**Marking criteria:**

(a) Substitute 3,25 and 108 in the formula  $n = \frac{m}{M}$  ✓

(b) Substitute  $6,02 \times 10^{23}$  in  $n(\text{e}^-) = \frac{N}{N_A}$  ✓

(c) Substitute 0,03 mol in  $n(\text{e}^-) = \frac{N}{N_A}$  ✓  
(Substitute 96 500 in formula  $Q = nF$ )

(d) Substitute 30(60) OR 1 800 ✓

(e) Final answer: 1,61 A ✓

**Nasienkriteria:**

(a) Vervang 3,25 en 108 in die formule  $n = \frac{m}{M}$  ✓

(b) Vervang  $6,02 \times 10^{23}$  in  $n(\text{e}^-) = \frac{N}{N_A}$  ✓

(c) Vervang 0,03 mol in  $n(\text{e}^-) = \frac{N}{N_A}$  ✓  
(Vervang 96 500 in formule  $Q = nF$ )

(d) Vervang 30(60) OF 1 800 ✓

(e) Finale antwoord: 1,61 A ✓

**OPTION 1/OPSIE 1:**

$$n(\text{Ag}) = \frac{m}{M}$$

$$= \frac{3,25}{108} \quad \checkmark \text{ (a)}$$

$$= 0,03 \text{ mol}$$

$$n(\text{e}^-) = \frac{N}{N_A}$$

$$\checkmark \text{ (c)} \quad 0,03 = \frac{N}{6,02 \times 10^{23}} \quad \checkmark \text{ (b)}$$

$$N \text{ e}^- = 1,81 \times 10^{22}$$

$$N \text{ e}^- = \frac{Q}{e} \quad \text{OF/OR} \quad \frac{Q}{q_e}$$

$$1,81 \times 10^{22} = \frac{Q}{1,6 \times 10^{-19}}$$

$$Q = 2\,889,6 \text{ C}$$

$$I = \frac{Q}{\Delta t}$$

$$= \frac{2\,889,6}{30(60)} \quad \checkmark \text{ (d)}$$

$$= 1,61 \text{ A} \quad \checkmark \text{ (e)}$$

**OPTION 2/OPSIE 2:**

$$n(\text{Ag}) = \frac{m}{M}$$

$$= \frac{3,25}{108} \quad \checkmark \text{ (a)}$$

$$= 0,03 \text{ mol} = n \text{ e}^-$$

$$\checkmark \text{ (b)} \quad Q = 0,03 \times 96\,500 \quad \checkmark \text{ (c)}$$

$$= 2\,895 \text{ C}$$

$$I = \frac{Q}{\Delta t}$$

$$= \frac{2\,895}{30(60)} \quad \checkmark \text{ (d)}$$

$$= 1,61 \text{ A} \quad \checkmark \text{ (e)}$$

(5)

[9]

TOTAL/TOTAAL:

150